Lecture 2.1: DOM Manipulation and Selectors

Instructor note: The page uses inline styles matching the template. For consistency across lectures consider linking to `notes-template.css`. Also ensure filenames and headings match your lecture numbering.

What is the DOM?

The **DOM (Document Object Model)** is the browser's in-memory representation of your HTML page. JavaScript can read and change the DOM to update what the user sees.

Goal of this lecture

Your lecture shows two important skills:

- **Select elements** from the page (find the node you want)
- **Update content** using properties like `innerText`

The HTML structure (from your example)

You used multiple headings and lists to practice selecting by tag, class, id, and a nested selector.

Old way: classic DOM selection methods

```
document.getElementsByTagName("h1")[0].innerText = "Using getElementsByTagName";
document.getElementsByClassName("diff")[0].innerText = "Using getElementsByClassName";
document.getElementById("special").innerText = "Using getElementById";
```

Output:

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```
// After running (DOM mutated):
<h1>Using getElementsByTagName</h1>
<h1 class="diff">Using getElementsByClassName</h1>
<h1 id="special">Using getElementById</h1>
```

What to notice

- `getElementById` returns a **single element**.
- `getElementsByTagName` and `getElementsByClassName` return a **collection**, so you used `[0]` to pick the first item.

New way: `querySelector` and CSS selectors

`querySelector` accepts a **CSS selector** and returns the **first matching element**.

```
document.querySelector("h1").innerText = "using querySelector('h1')";
document.querySelector(".diff").innerText = "using querySelector('.diff')";
document.querySelector("#special").innerText = "using querySelector('#special')";
document.querySelector(".first > .diff2").innerText = "changed via JS";
```

Output:

```
// After running (DOM mutated):
<h1>using querySelector('h1')</h1>
<h1 class="diff">using querySelector('.diff')</h1>
<h1 id="special">using querySelector('#special')</h1>
<li class="diff2">changed via JS</li> (first list)
```

Selector	Meaning	Matches
`h1`	Tag selector	All `<h1>` elements (first one returned by `querySelector`)

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Selector	Meaning	Matches
`.diff`	Class selector	Element with class `diff`
`#special`	ID selector	Element with id `special`
`.first > .diff2`	Child selector	Direct child with class `diff2` inside `.first`

> **Important:** If you want *all* matches, use `querySelectorAll` (it returns a NodeList).

`innerText` vs `innerHTML` (extra insight)

- `innerText` : sets/gets plain text (safe for user content).
- `innerHTML` : sets HTML markup (powerful, but can be unsafe if you insert untrusted text).

```
// Text only
document.querySelector("p").innerText = "Hello <b>world</b>";

// HTML (renders the bold tag)
document.querySelector("p").innerHTML = "Hello <b>world</b>";
```

Output:

```
// After innerText: shows the literal text "Hello <b>world</b>"
// After innerHTML: shows Hello world with the word "world" bolded
```

Reading values

Your lecture reads the paragraph text:

```
console.log(document.querySelector("p").innerText);
```


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Output:

```
Sample text
```

Common mistakes

- Calling selectors before the element exists (place scripts at the bottom or use `DOMContentLoaded`).
- Forgetting `.` for class and `#` for id in `querySelector`.
- Expecting `querySelector` to return multiple elements (it returns only the first match).

Practice tasks

1. Change the text of the **second** `h1` using `querySelectorAll`.
2. Change all `li` items with class `diff2` to "Updated".
3. Add a new `li` item at the end of the first list using JavaScript.

JS Refined Notes — Lecture 2.1

> **Instructor insight / accuracy checks:** Notes: - Use `querySelectorAll` when you need all matches (returns a NodeList). - Place scripts after DOM or use `DOMContentLoaded` to avoid selecting missing elements. - Prefer `textContent` for faster reads/writes when you don't need layout-aware behavior. - Consider linking to `notes-template.css` to avoid inline style duplication across files.